ON A REGULAR NOETHERIAN SEPARATED SPACE EVERY INVERTIBLE MODULE IS A DIFFERENCE OF TWO EFFECTIVE CARTIER DIVISORS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1843-1852 of sources/stacks-spaces-divisors/pinned-original-source.tex; proof span is lines 1854-1877, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool spaces-divisors wave35 (sd02 batch).

Editorial note. rqual sv-mismatch fix 2026-10-14: source_version aligned to SOURCE.json (semicolon+git form, intake authority; package evidence.json already correct); revision bumped for re-assembly per publication rotation (takeover-line defer ticket, relayed by the s15 line).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 01YB: sources/stacks-spaces-divisors/pinned-original-source.tex, source lines 1877–1877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0847: sources/stacks-spaces-divisors/pinned-original-source.tex, source lines 1735–1744. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06NH: sources/stacks-spaces-divisors/pinned-original-source.tex, source lines 1877–1877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a regular Noetherian separated algebraic space over S. Then every invertible $\mathcal{O}_X$-module is isomorphic to*

$$\mathcal{O}_X(D - D') = \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')^{\otimes -1}$$

for some effective Cartier divisors D, D' in X.

Date: stacks-project:spaces-divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Choose a dense affine open $U \subset X$ such that $\mathcal{L}|_U$ is trivial. This is possible because X has a dense open subspace which is a scheme, see Properties of Spaces, Proposition 06NH. Denote $s : \mathcal{O}_U \to \mathcal{L}|_U$ the trivialization. The complement of U is an effective Cartier divisor D. We claim that for some $n > 0$ the map s extends uniquely to a map

$$s : \mathcal{O}_X(-nD) \longrightarrow \mathcal{L}$$

The claim implies the lemma because it shows that $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{O}_X(nD)$ has a regular global section hence is isomorphic to $\mathcal{O}_X(D')$ for some effective Cartier divisor D' by Lemma 0847. To prove the claim we may work étale locally. Thus we may assume X is an affine Noetherian scheme. Since $\mathcal{O}_X(-nD) = \mathcal{I}^n$ where $\mathcal{I} = \mathcal{O}_X(-D)$ is the ideal sheaf of D in X, this case follows from Cohomology of Schemes, Lemma 01YB. □